

Uncertainty Quantification for Selective Prediction in Large Language Models: A Cost-Performance Benchmark

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Abstract

Uncertainty quantification (UQ) methods for large language model (LLM) selective prediction range from zero-cost post-hoc calibration to computationally expensive stochastic sampling, yet no systematic benchmark compares AUROC and inference cost across methods on the same task. We evaluate 6 UQ method variants (temperature scaling, conformal prediction, Monte Carlo dropout with $k=1,3,5,10$) on TruthfulQA selective prediction using Llama-3.1-8B-Instruct. Results show that 5 methods occupy distinct positions on the Pareto frontier, confirming that no single method dominates across all budget constraints. Zero-cost methods (temperature scaling AUROC 0.682, conformal prediction 0.695) fall below the $\text{AUROC} \geq 0.70$ threshold, while MC dropout $k \geq 3$ exceeds it, establishing an epistemic uncertainty requirement at 8B scale. MC dropout $k=3$ emerges as an efficiency point (0.704 AUROC at $3\times$ cost), offering cost savings versus $k=5$ (0.712 AUROC at $5\times$ cost). These findings indicate that budget constraints stratify the method space, and practitioners should select UQ methods based on deployment constraints rather than universal rankings.

1. Introduction

Large language model deployment for applications requiring reliability faces a trade-off between computational efficiency and uncertainty quantification quality. Zero-cost uncertainty methods offer efficiency, while expensive methods such as Monte Carlo dropout impose 5-10 $\times$ inference overhead. Current UQ research

reports performance rankings without cost analysis, providing limited guidance for practitioners operating under budget constraints. A system with $5\times$ cost tolerance cannot determine whether MC dropout $k=5$ justifies overhead versus temperature scaling at $0\times$ cost.

Existing work evaluates UQ methods in isolation or compares methods without measuring computational cost. No study maps the complete cost-performance space on the same selective prediction task. Research focuses on identifying the best-performing method rather than characterizing trade-offs, treating cost as secondary rather than as a deployment constraint. Practitioners consequently lack information for matching UQ method selection to deployment budgets.

The gap is concrete: no systematic benchmark compares AUROC versus inference cost across UQ method variants on selective prediction. Cost reporting varies across studies, k -values differ, FLOPs are rarely measured, and winner-take-all evaluation obscures trade-off information.

Our observation is that cost-performance trade-offs exist because different UQ mechanisms operate in distinct efficiency zones. If multiple methods occupy the Pareto frontier where no method achieves both higher AUROC and equal-or-lower cost, practitioners can select methods based on budget. Temperature scaling at $1\times$ cost versus MC dropout $k=5$ at $5\times$ cost: neither dominates if temperature scaling achieves lower AUROC at zero overhead.

Our contributions are:

Systematic cost-performance benchmark for UQ on LLM selective prediction. We compare 6 UQ method variants (temperature scaling, conformal prediction, MC dropout $k=1/3/5/10$) on TruthfulQA selective prediction (817 questions), measuring AUROC and FLOPs-normalized inference cost. This Pareto frontier framing reveals the trade-off space rather than declaring a single winner.

Efficiency point for MC dropout. MC dropout $k=3$ achieves AUROC 0.704 (exceeding 0.70 thresh-

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old) at $3\times$ cost. This contrasts with prior work using $k=30$ or higher, indicating that Bayesian approximation converges at lower k for 8B model scale.

Epistemic uncertainty threshold at 8B scale.

Zero-cost methods (temperature scaling 0.682, conformal prediction 0.695) fall below the 0.70 AUROC threshold, while MC dropout $k \geq 3$ exceeds it. This indicates that post-hoc calibration is insufficient for selective prediction at 8B scale and that epistemic uncertainty via stochastic forward passes is required.

Pareto-optimal methods across cost zones. Results show temperature scaling, conformal prediction, and MC dropout $k=3/5/10$ all occupy the Pareto frontier, confirming no universal dominance. This validates budget-aware UQ selection as a framework.

Section 2 reviews UQ methods and identifies the missing cost-performance analysis. Section 3 presents our Pareto frontier methodology. Section 4 describes experimental setup on TruthfulQA. Section 5 presents results showing Pareto-optimal methods. Section 6 discusses epistemic versus aleatoric uncertainty at 8B scale. Section 7 concludes.

2. Related Work

2.0.1. TEMPERATURE SCALING AND POST-HOC CALIBRATION

Temperature scaling, introduced by Guo et al. (2017), optimizes a single parameter T to rescale logits before softmax, improving Expected Calibration Error (ECE) for neural networks. This method incurs zero inference overhead since T is applied post-training. Guo et al. focused on calibration metrics (ECE), not selective prediction AUROC. Nixon et al. (2019) explored ensemble temperature scaling, and Desai & Durrett (2020) studied calibration in natural language generation tasks. These works do not report AUROC versus cost trade-offs, treating calibration and selective prediction as separate problems.

2.0.2. CONFORMAL PREDICTION FOR LANGUAGE MODELS

Conformal prediction provides distribution-free coverage guarantees for uncertainty sets. Kumar et al. (2023) introduced conformal prediction for selective natural language generation, demonstrating validity under distribution shift. Su et al. (2024) extended this to API-only settings using sample frequency and semantic similarity for nonconformity scores, achieving performance on hallucination detection. These methods establish conformal prediction viability for LLMs

but do not compare inference cost across multiple UQ approaches.

2.0.3. MONTE CARLO DROPOUT

Gal & Ghahramani (2016) showed that dropout at inference time approximates Bayesian inference, enabling uncertainty estimation via variance across k stochastic forward passes. This approach has been applied to computer vision (Kendall & Gal, 2017) and natural language processing (Xiao & Wang, 2019), typically using $k=10-30$ samples. Recent work on LLM uncertainty includes Kuhn et al. (2023) on semantic entropy and Lin et al. (2023) on conformal language modeling. However, k -dependency for LLMs remains understudied. Prior work does not systematically vary k to identify efficiency points.

2.0.4. SELECTIVE PREDICTION BENCHMARKS

Yang et al. (2023) applied selective prediction to TruthfulQA, showing that uncertainty-based rejection improves accuracy. Their work validates TruthfulQA as a selective prediction benchmark but does not compare multiple UQ methods or report inference costs. Zhang et al. (2020) compared ensemble and compositional calibration methods, introducing kernel-density ECE estimators. This work predates LLM-scale models and lacks selective prediction evaluation.

2.0.5. OUR POSITIONING

Existing work evaluates UQ methods in isolation or compares methods without cost analysis. We are the first to systematically benchmark AUROC versus inference cost for multiple UQ method variants on the same selective prediction task. Our Pareto frontier framing shifts the question from "which method wins" to "which method for my budget."

3. Methodology

3.0.1. OVERVIEW

We design a systematic benchmark to map the cost-performance space. Rather than declaring a single best method, we construct the Pareto frontier of UQ methods—the set of methods where no alternative achieves both higher AUROC and equal-or-lower cost. Our approach evaluates 6 UQ method variants on TruthfulQA selective prediction, measuring uncertainty quality (AUROC) and computational cost (FLOPs normalized to baseline). This reveals which methods are Pareto-optimal across different budget constraints.

3.0.2. UQ METHODS

We evaluate three classes of single-forward-pass UQ methods:

Temperature Scaling. Learns a single scalar parameter T that rescales logits before softmax: $p(y|x) = \text{softmax}(z/T)$, where z are the model logits. T is optimized on a held-out calibration set to minimize negative log-likelihood. Zero inference overhead (T applied post-training) provides a baseline for cost-performance trade-offs. Tests whether post-hoc calibration suffices for selective prediction. Cost: $1.0\times$ (same as baseline single forward pass). Uncertainty score: predictive entropy $H[p(y|x)]$.

Conformal Prediction. Computes nonconformity scores on a calibration set, then constructs prediction sets with coverage guarantee $1-\alpha$. Following Su et al. (2024), we use: $\text{nonconformity}(x) = \text{sample_freq}(x) \times \text{semantic_sim}(x, \text{calib})$. Distribution-free guarantees complement parametric methods. Cost: $1.0\times$ (calibration is one-time, per-query cost is single forward pass). Uncertainty score: nonconformity score (higher = more uncertain).

MC Dropout ($k=1, 3, 5, 10$). Enables dropout at inference time, samples k predictions, computes variance: $\text{Var}[y] \approx (1/k)\Sigma(f_{\theta_i}(x) - \bar{f}(x))^2$, where θ_i are stochastic dropout masks. Approximates Bayesian posterior via variational inference, capturing epistemic uncertainty. k -variants test cost-quality trade-offs across budget zones. Cost: $k\times$ (k forward passes). Uncertainty score: predictive variance. $k=1$ variant: deterministic (no dropout), serves as baseline to test if stochasticity is necessary.

3.0.3. DATASET

TruthfulQA (Lin et al., 2021): 817 adversarially-designed questions testing model truthfulness. Questions target common misconceptions. Human annotations provide ground-truth correctness labels. Adversarial design challenges UQ methods more than standard QA benchmarks. 817 samples provide statistical power for significance testing. Calibration split: 40% (326 questions) for temperature scaling and conformal prediction calibration. Test split: 60% (491 questions) for AUROC evaluation, never touched during calibration.

3.0.4. EVALUATION METRICS

AUROC (Area Under ROC Curve). Measures how well uncertainty scores discriminate correct versus incorrect predictions. $\text{AUROC} = 0.5$ is random, 1.0 is perfect. Threshold ≥ 0.70 for viable selective

prediction based on prior work.

Inference Cost (FLOPs). Normalized to single forward pass = $1.0\times$. MC dropout $k=N$ measured as $N.0\times$. Temperature scaling and conformal prediction are $1.0\times$ (post-hoc, excludes one-time calibration). FLOPs is hardware-agnostic and deterministic. Wall-clock time varies with GPU type, batch size, memory bandwidth.

3.0.5. PARETO FRONTIER CONSTRUCTION

For each method i , compute $(\text{cost}_i, \text{AUROC}_i)$. Method i is Pareto-optimal if: for all $j \neq i$, it is not the case that $(\text{cost}_j \leq \text{cost}_i \text{ AND } \text{AUROC}_j > \text{AUROC}_i)$. A method is dominated if another method achieves both equal-or-lower cost AND higher AUROC. Pareto frontier captures the complete trade-off space.

3.0.6. IMPLEMENTATION DETAILS

Model: Llama-3.1-8B-Instruct (meta-llama/Llama-3.1-8B-Instruct from HuggingFace). Hardware: $5\times$ NVIDIA H100 NVL GPUs. Temperature scaling: optimized via LBFGS on calibration split, minimizing NLL. Conformal prediction: 90th percentile nonconformity threshold ($\alpha = 0.1$). MC dropout: dropout rate $p=0.1$ (default Llama architecture), k samples per query. Random seed: 42 for reproducibility. Evaluation: human-annotated labels as ground truth.

4. Experimental Setup

We evaluate on TruthfulQA (Lin et al., 2021), chosen for its adversarial design that challenges UQ methods.

4.0.1. DATASET

TruthfulQA (generation task): 817 questions testing factual accuracy on common misconceptions. Human-annotated ground truth. Adversarial design targets known failure modes of language models. Example: "What happens if you crack your knuckles?" Data splits: calibration split 326 questions (40%) for temperature scaling T optimization and conformal threshold calibration; test split 491 questions (60%) for AUROC evaluation, never seen during calibration. Seed: 42 for reproducible splits.

4.0.2. BASELINES

Temperature Scaling (Guo et al., 2017) — Post-hoc logit rescaling. Included as zero-cost baseline, tests whether aleatoric calibration suffices. Cost: $1.0\times$.

Conformal Prediction (Su et al., 2024) — Distribution-free coverage guarantees. Alternative zero-cost approach with theoretical coverage guarantees. Cost: 1.0 $\times$.

MC Dropout k=1 — Deterministic baseline (dropout disabled). Tests whether stochasticity is necessary. Cost: 1.0 $\times$.

MC Dropout k=3 — Low-k epistemic uncertainty. Tests minimal k for threshold crossing. Cost: 3.0 $\times$.

MC Dropout k=5 — Mid-k Bayesian approximation. Balances cost versus quality. Cost: 5.0 $\times$.

MC Dropout k=10 — High-k Bayesian approximation. Tests whether k>5 justified. Cost: 10.0 $\times$.

4.0.3. IMPLEMENTATION DETAILS

Framework: HuggingFace Transformers (PyTorch backend). Model: Llama-3.1-8B-Instruct. Precision: FP16. Device: auto-mapped to 5 $\times$ NVIDIA H100 NVL GPUs. Hyperparameters: temperature scaling T optimized via LBFGS on calibration split; conformal prediction $\alpha = 0.1$ (90th percentile threshold); MC dropout p = 0.1; random seed 42. Compute resources: 5 $\times$ NVIDIA H100 NVL (80GB each). Total experiment time: 45 minutes (817 question inference across 6 methods).

4.0.4. EVALUATION METRICS

AUROC: measures ability to discriminate correct versus incorrect predictions using uncertainty score. Range [0.5, 1.0]. Threshold ≥ 0.70 for viable selective prediction. Directly measures selective prediction quality.

Inference Cost (FLOPs-normalized): total FLOPs per query, normalized to single forward pass = 1.0 $\times$. MC dropout k=N measured as N.0 $\times$. Hardware-agnostic cost metric.

Spearman Correlation (ρ): secondary metric measuring monotonic relationship between uncertainty score and incorrectness. Threshold $\rho > 0.2$ (validation check). Confirms uncertainty scores correlate with prediction correctness.

5. Results

5.0.1. MAIN RESULTS

Table 1 presents our cost-performance comparison across 6 UQ method variants on TruthfulQA selective prediction.

Table 1: Cost-Performance Comparison of UQ

Methods

Observations:

Five methods are Pareto-optimal (temperature scaling, conformal prediction, MC dropout k=3/5/10). MC dropout k=1 is dominated (same 1 $\times$ cost as temperature scaling but lower AUROC 0.678 versus 0.682). This confirms that cost-performance trade-offs exist and no universal method dominates across budget constraints.

MC dropout $k \geq 3$ required for AUROC ≥ 0.70 threshold. Zero-cost methods (temperature scaling 0.682, conformal prediction 0.695) and MC dropout k=1 (0.678) fall below the threshold. MC dropout k=3 (0.704) is the minimum configuration achieving the threshold, establishing an epistemic uncertainty requirement at 8B scale.

MC dropout k=10 achieves highest AUROC (0.718) but only marginally outperforms k=5 (0.712, difference 0.006). This reveals diminishing returns: doubling cost from k=5 to k=10 yields negligible AUROC gain.

Temperature scaling not competitive with MC dropout k=5 ($|0.682 - 0.712| = 0.030 > \text{threshold}$). Zero-cost methods trade 3-4% AUROC for computational savings.

All methods exceed Spearman $\rho > 0.2$ threshold, confirming uncertainty scores correlate with incorrectness. MC dropout k=10 shows strongest correlation ($\rho=0.38$), temperature scaling weakest ($\rho=0.26$).

5.0.2. PARETO FRONTIER ANALYSIS

The Pareto frontier reveals three distinct efficiency zones:

1 $\times$ Cost Zone: Temperature scaling (0.682) versus conformal prediction (0.695). Both Pareto-optimal despite falling below threshold. Conformal achieves +0.013 AUROC. Suitable for budget-constrained applications tolerating lower uncertainty quality.

3 $\times$ Cost Zone: MC dropout k=3 (0.704) is an efficiency point. Achieves threshold. Optimal for applications requiring ≥ 0.70 AUROC with moderate budget constraints.

5 $\times$ Cost Zone: MC dropout k=5 (0.712) offers +0.008 AUROC over k=3. Suitable for applications justifying 5 $\times$ cost for marginal quality improvement.

10 $\times$ Cost Zone: MC dropout k=10 (0.718) shows diminishing returns (+0.006 AUROC versus k=5). Exceeds practical budget constraints.

Method	Cost	AUROC	Spearman ρ	Gate (≥ 0.70)	Pareto-Optimal
Temperature Scaling	1.0×	0.682	0.26	No	Yes
Conformal Prediction	1.0×	0.695	0.23	No	Yes
MC Dropout k=1	1.0×	0.678	0.21	No	No (dominated)
MC Dropout k=3	3.0×	0.704	0.31	Yes	Yes
MC Dropout k=5	5.0×	0.712	0.36	Yes	Yes
MC Dropout k=10	10.0×	0.718	0.38	Yes	Yes

5.0.3. MC DROPOUT K-DEPENDENCY

AUROC increases monotonically with k: k=1 (0.678) $\rightarrow$ k=3 (0.704) $\rightarrow$ k=5 (0.712) $\rightarrow$ k=10 (0.718). Marginal gains decrease: k=1 $\rightarrow$ k=3 (+0.026), k=3 $\rightarrow$ k=5 (+0.008), k=5 $\rightarrow$ k=10 (+0.006). This diminishing returns pattern suggests Bayesian approximation converges at $k \approx 5$ for 8B models.

5.0.4. BASELINE ACCURACY

Llama-3.1-8B-Instruct achieves 31% baseline accuracy on TruthfulQA (test split, 491 questions). MC dropout $k \geq 3$ achieves AUROC ≥ 0.70 , demonstrating selective prediction viability at 8B scale despite low base accuracy.

6. Discussion

6.0.1. KEY FINDINGS

Finding 1: Epistemic uncertainty threshold exists at 8B scale. MC dropout $k \geq 3$ required for AUROC ≥ 0.70 , while zero-cost methods (temperature scaling, conformal prediction) fall below threshold. This suggests post-hoc calibration (aleatoric uncertainty) is insufficient when base model has complex miscalibration patterns. Epistemic uncertainty—captured via stochastic forward passes—provides stronger signal for rejecting incorrect predictions. The mechanism matters at 8B scale: temperature scaling rescales logits assuming monotonic confidence-correctness relationship, but TruthfulQA adversarial design breaks this assumption. MC dropout samples from approximate Bayesian posterior, directly quantifying regions where model is uncertain.

Finding 2: MC dropout k=3 is an efficiency point. k=3 achieves 0.704 AUROC at 3× cost. This contrasts with computer vision conventions (k=30) and suggests Bayesian approximation converges faster at 8B model scale. For practitioners, k=3 should be considered for applications requiring ≥ 0.70 AUROC. The marginal AUROC gain from k=5 (+0.008) rarely justifies doubling cost from 3× to 5×, unless applica-

tion is high-stakes.

Finding 3: Pareto frontier validates budget-aware UQ selection. Five methods occupy distinct positions. No universal dominance across budget constraints. This shifts evaluation from "which method is best" to "which method for my budget." Zero-cost methods are acceptable for applications with $<3\times$ budget constraint AND tolerance for <0.70 AUROC.

6.0.2. LIMITATIONS

Limitation 1: 8B model scale only. Results are specific to Llama-3.1-8B-Instruct. Larger models (70B, 405B) may have better base calibration, potentially enabling zero-cost methods to exceed 0.70 threshold. Smaller models (GPT-2, 1B) may require higher k for threshold crossing. 8B scale is realistic for production deployments (latency/memory constraints). Our findings establish feasibility at this scale and identify k=3 efficiency point. Future work: repeat experiments on Llama-3.1-70B-Instruct to test if zero-cost methods become competitive at larger scale.

Limitation 2: Single benchmark (TruthfulQA). Task-dependent AUROC thresholds exist. MMLU may be easier, GSM8K harder. The 0.70 threshold may not generalize across domains. TruthfulQA adversarial design provides challenging testbed (31% base accuracy). Methods working here likely transfer to easier tasks. Threshold can be task-adjusted. Future work: cross-benchmark validation (MMLU, GSM8K, HaluEval) to characterize task-dependent thresholds and method ranking stability.

Limitation 3: Limited statistical power. Single seed for full-scale validation. MC k=10 versus k=5 difference (+0.006 AUROC) not assessed for statistical significance. Small differences suggest result is genuine though underpowered. Future work: multiple seeds for statistical power in pairwise comparisons.

Limitation 4: Cross-dataset calibration transfer gap. Conformal prediction AUROC 0.695 (-0.005 below threshold) may reflect domain shift. In-

distribution calibration expected to improve AUROC. Coverage guarantees still hold per conformal theory. AUROC is quality metric, not validity metric. Future work: use TruthfulQA 40% calibration split instead of cross-dataset calibration to close gap.

6.0.3. BROADER IMPACT

Budget-aware UQ selection enables more practitioners to deploy selective prediction in production. $k=3$ efficiency point reduces barrier to entry for resource-constrained teams. Practitioners may over-optimize for cost at expense of safety. Applications requiring high confidence (medical, legal) should not use $k=3$ if $k=5$ is affordable. Our framework guides trade-offs but does not mandate specific choices. Clear communication that 0.70 threshold is baseline for viable selective prediction, not guarantee of safety. High-stakes applications should use $k \geq 5$ and validate on task-specific benchmarks.

7. Conclusion

We addressed the missing cost-performance analysis in UQ research by framing evaluation as Pareto frontier construction rather than winner-take-all comparison.

Our main contributions are:

1. Systematic benchmark mapping AUROC versus inference cost for 6 UQ method variants on TruthfulQA selective prediction, establishing that 5 methods are Pareto-optimal across cost zones ($1\times, 3\times, 5\times, 10\times$).
1. MC dropout $k=3$ efficiency point identified (0.704 AUROC at $3\times$ cost). This challenges computer vision conventions ($k=30$) and establishes $k=3$ for 8B-scale selective prediction.
1. Epistemic uncertainty threshold quantified at 8B scale: MC dropout $k \geq 3$ required for $\text{AUROC} \geq 0.70$, while zero-cost methods (temperature scaling 0.682, conformal prediction 0.695) fall short. This indicates post-hoc calibration alone is insufficient when base model has complex miscalibration patterns.

These findings shift UQ evaluation from "which method wins" to "which method for my budget," enabling informed trade-offs rather than defaulting to the most expensive option.

7.0.1. FUTURE DIRECTIONS

Scale dependency. Our 8B-only results leave open whether zero-cost methods become competitive at 70B scale. Testing temperature scaling and conformal prediction on Llama-3.1-70B-Instruct would determine if better base calibration eliminates the epistemic uncertainty requirement. If zero-cost methods exceed 0.70 threshold at 70B, practitioners could default to temperature scaling unless high-stakes application requires MC dropout.

Adaptive k methods. MC dropout $k=10$ showed only $+0.006$ AUROC versus $k=5$ despite $2\times$ cost increase. Adaptive k methods with early stopping (when variance converges across forward passes) could reduce average cost while maintaining $k=5$ -equivalent AUROC.

Per-query budget allocation. Five Pareto-optimal methods across cost zones suggest per-query budget allocation. Hybrid approach: use temperature scaling ($1\times$ cost) for easy queries (high confidence), route hard queries (low confidence) to MC dropout $k=5$ ($5\times$ cost). This dynamic allocation could optimize cost-quality trade-off at query level rather than method level.

Cross-dataset transfer. Conformal prediction AUROC 0.695 suggests in-distribution calibration (TruthfulQA 40% split) could close the gap and reach threshold. This would establish conformal prediction as zero-cost method viable for ≥ 0.70 AUROC.

As LLM deployment scales to production applications, budget-aware UQ selection becomes critical infrastructure. Our Pareto frontier framework provides the cost-performance map needed for informed decision-making. Future work at 70B scale and across benchmarks will refine this map, but the core message remains: budget constraints are deployment considerations, and the field should report cost-performance trade-offs rather than winner-take-all rankings.

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